Kyl Center for Water Policy at Morrison Institute

Arizona State University

From Copper, Cattle and Cotton to Chips and Cloud Computing: Large Water Uses in Central Arizona

“AI will open up new ways of doing things that we cannot even imagine today.”

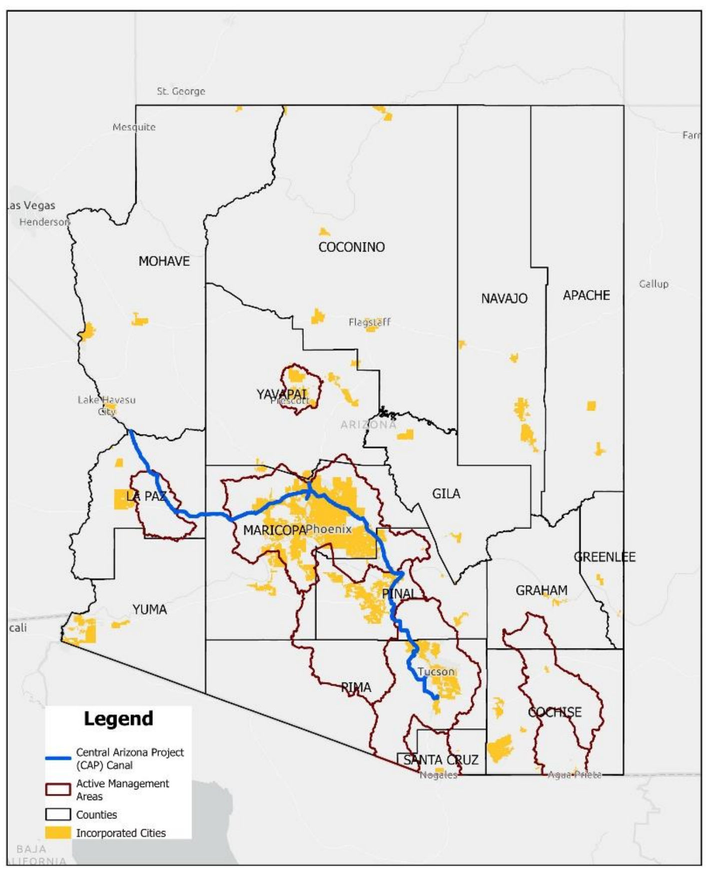
— *Sundar Pichai, CEO of Google*

“America is the country that started the AI race. And as President of the United States, I'm here today to declare that America is going to win it.”

— *President Donald J. Trump, July 23, 2025*

<u>Study Area</u>

This report focuses on large water uses for non-agricultural purposes in the Phoenix, Pinal and Tucson Active Management Areas (AMAs). AMAs are regions where groundwater use is tracked and highly regulated pursuant to Arizona’s 1980 Groundwater Management Act (GMA). Throughout this report, the three AMA study area is referred to as “Central Arizona,” consistent with the nomenclature of the Central Arizona Water Conservation District and the Central Arizona Project canal, through which Colorado River water is imported into the region.



Introduction

A half-century ago, Arizona school children were taught that the pillars of Arizona's economy and growth were the Three Cs: copper, cattle and cotton. Over time, the state's economy has diversified, especially in Central Arizona, where around 85% of the state's population resides and where the municipal population has doubled since 1980. Mining copper, growing alfalfa to feed cattle and growing cotton are arguably water-intensive activities. Yet the development of at least two new "Cs"—chips, as in semi-conductor chips, and cloud, as in servers, databases and software that are accessed via the internet, have created consternation over water use by these newer and rapidly growing economic sectors.

To address these concerns and help inform community decision-making, this report explores the impacts of water demand from these new Cs and other industrial users and contextualizes that demand with respect to other large water uses that have long been woven into the pattern of urban water use. People within Arizona and beyond rely on the products, services and jobs provided by large water users. In Central Arizona, large water users include food processors, miners of sand and gravel to make concrete, beverage facilities such as bottling companies and breweries, turf facilities such as public parks, schools, sports complexes and golf courses, microchip manufacturers, power utilities and data centers, among others.

Here we explore large water uses in Central Arizona with a special focus on data centers. In Part 1 we look at the big picture of water uses and sources. In Part 2 we discuss water sources for urban uses. In Part 3 we examine large water uses within a municipal tap water system. In Part 4 we discuss considerations related to water use by data centers and in Part 5 we look at future trends.

This report relies on two primary data sources: the Arizona Department of Water Resources (ADWR) AMA demand and supply data, which quantifies annual demand for various use categories and sub-categories from 1985-2024, and the Kyl Center's 2022 Community Water Use study (CWU), an analysis of all physical sources and uses of water within community water systems in Central Arizona. In the data analysis for this report, we changed the categorization of ADWR AMA demand and supply data for cattle feedlots and dairy from industrial to agricultural, data for golf courses, parks, schools, cemeteries and HOAs from industrial to landscape & turf, and data for subdivision irrigation from municipal to landscape & turf. ADWR sector data related to lost & unaccounted for water was excluded. The 2022 CWU was carefully crafted from ADWR's database of groundwater pumping, the annual reports submitted by water users to ADWR, aerial maps, municipal water provider consumer confidence reports, direct communication with municipal water providers and other sources. This analysis is limited to water use across sectors for non-tribal uses. Most Tribal water use in Central

Arizona is for agricultural purposes; however, Tribes, which are sovereign, do not report water use to the state.

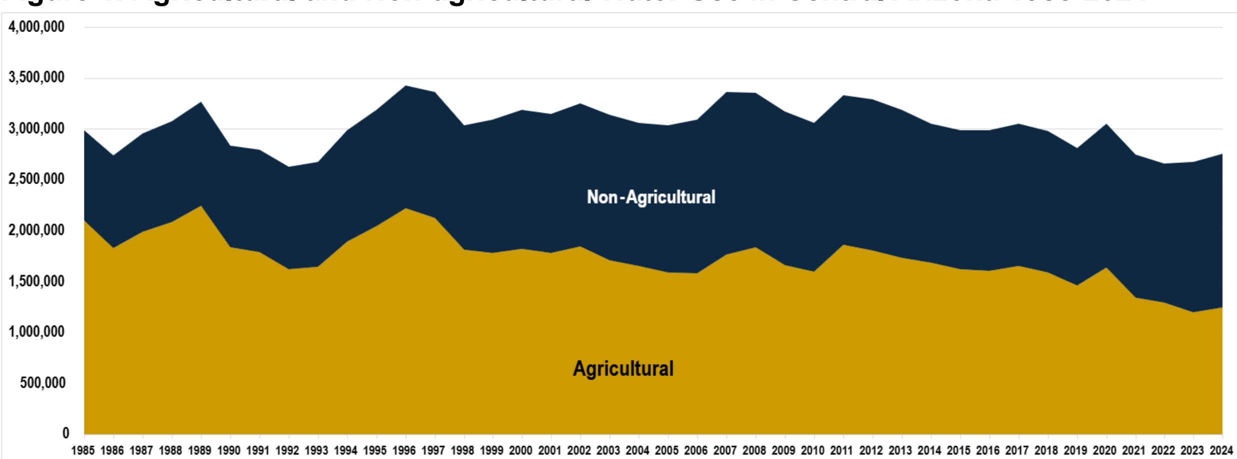
<u>Background</u>

Central Arizona has urbanized over the last several decades, and its economy has changed accordingly. Because much urban growth occurred on previously farmed lands and urban water uses are generally lower per-acre than agricultural ones, total non-tribal water use within Central Arizona has declined by around 9% over the last four decades as agriculture's contribution to the regional economy has diminished and other industries have emerged and grown.[1]

Table 1: Declines in Non-Tribal Water Use Since 1985

Phoenix AMA	**5%**
Pinal AMA	**16%**
Tucson AMA	**2%**

Figure 1: Agricultural and Non-agricultural Water Use in Central Arizona 1985-2024



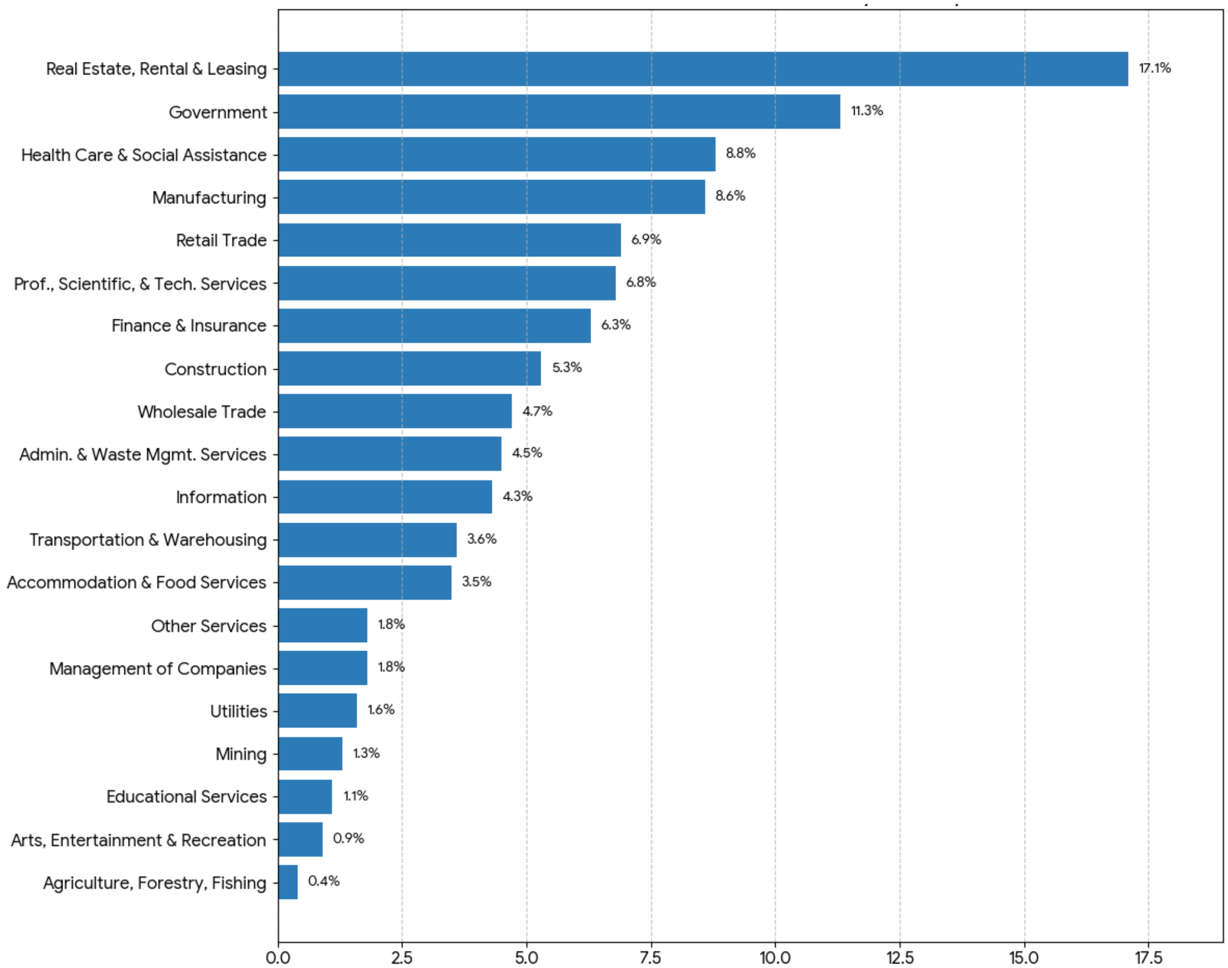
Today, agriculture represents less than 1% of the state's gross domestic product; mining, less than 2%. Instead, real estate, government and health care and social assistance lead the pack. As of 2023, manufacturing came in at number four. We can expect manufacturing to contribute even more to the regional economy as the Taiwan Semiconductor Manufacturing Company facility in north Phoenix continues to expand in conjunction with its many suppliers. This facility, alongside the expansion of Intel in Chandler and the emergence of data centers across Central Arizona, is driving a shift to a high-tech economy.

[1] Ariz. Dept. of Water Resources, *AMA Data (AMA Annual Supply and Demand Dashboard),* https://www.azwater.gov/ama/ama-data. (hereafter ADWR, AMA Data).

Table 2: Ratio of Non-Farm to Farm Jobs[2]

	1985	2020
Maricopa County	150:1	395:1
Pinal County	5:1	19:1
Pima County	80:1	100:1

Table 3: Arizona's 2023 Gross Domestic Product by Industry[3]



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[2] U.S. Bureau of Economic Analysis (BEA) Table CAEMP25N; Arizona Office of Economic Opportunity (2023-2024 data); MAP Dashboard (University of Arizona); University of Arizona Cooperative Extension & Arizona Farm Bureau (historical data).
[3] Steven L. Byers and Glenn Farley, *Arizona's 2023 Economic Performance Index*, Common Sense Inst., February 2024, https://www.commonsenseinstituteus.org/arizona/research/jobs-and-our-economy/arizonas-2023-economic-performance-index.

With this shift have come concerns about water uses – concerns that may be intensified by current challenges to the region's water supplies. The flows of one important water supply, the Colorado River, are in decline[4] and in the Phoenix and Pinal AMAs, the state has restricted some forms of groundwater reliant growth.[5] It is more important than ever for local elected officials and other stakeholders to determine whether a proposed economic enterprise—high-tech or otherwise—is worth the water.

Part I. Water Use and Water Supply – the Big Picture

Relative Water Uses by Sector

While non-agricultural demand has gradually increased, in Central Arizona agriculture continues to use more water than any other sector.

Agricultural Water Use

Given the region's long growing season and arid climate, agricultural users can easily use 6 acre-feet per acre per year or more depending on cropping patterns.[6] In contrast, annual water use per acre in urban areas in Central Arizona ranges from around 0.2 acre-feet per acre to 2.6 acre-feet per acre per year and averages about one acre-foot per acre per year.[7]

Water use for all non-agricultural sectors combined did not match agricultural uses until 2021. In 2024, agriculture used approximately 1.24 million acre-feet out of 2.75 million acre-feet in total.[8]

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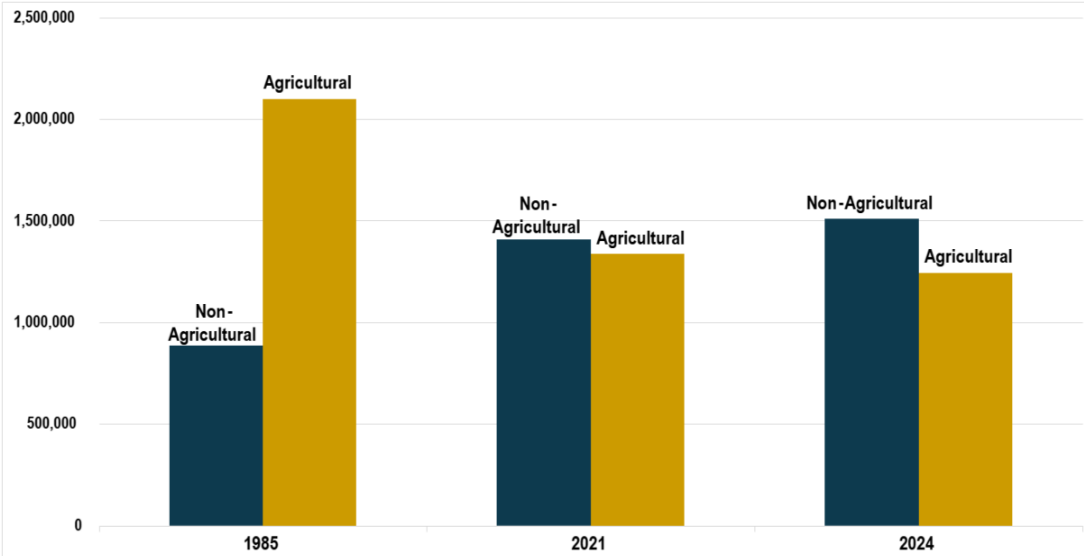
[4] Jack Schmidt et al., *The Colorado River water crisis: Its origin and the future*, WIREs Water,2023, https://doi.org/10.1002/wat2.1672.
[5] Kyl Center for Water Policy, *New Phoenix AMA Model Shows Limits of Groundwater as an Assured Water Supply*, ASU Morrison Inst. for Public Policy, June 2023, https://morrisoninstitute.asu.edu/publication/new-phoenix-ama-model-shows-limits-groundwater-assured-water-supply.
[6] *See* Arizona Department of Water Resources, *Phoenix Active Management Area 5th Management Plan* § 4.2.1.2 (calculation of irrigation water duties).
[7] Kyl Center for Water Policy, *Water and Land Use in Central Arizona*, Arizona Water Blueprint, ASU Morrison Inst. for Public Policy, October 8, 2025, https://storymaps.arcgis.com/stories/eb51183c1305428187c699d004cbf77b.
[8] ADWR, AMA Data

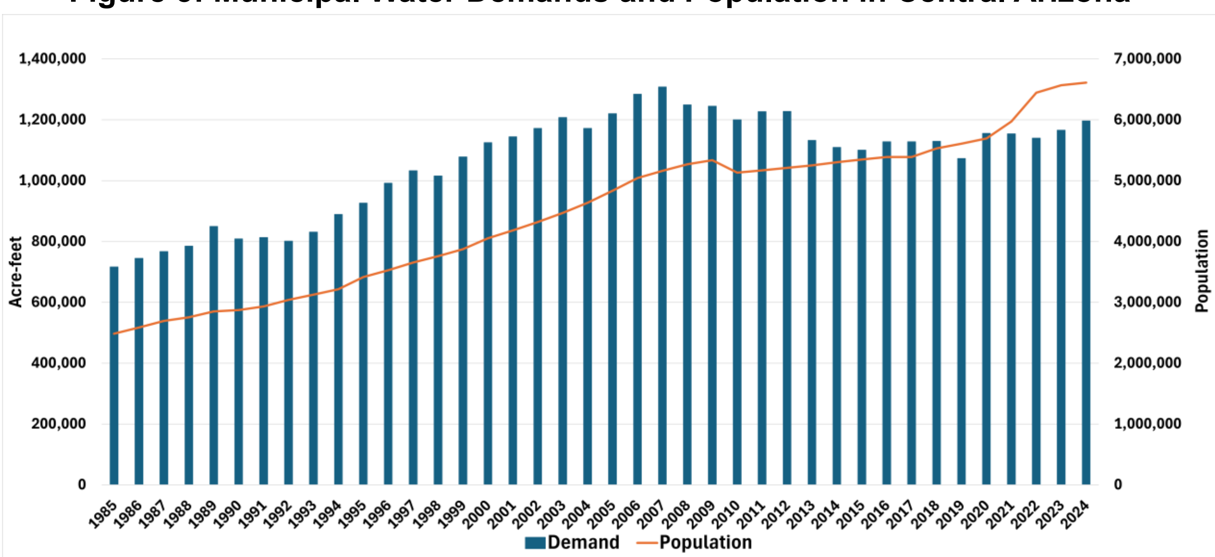
Figure 2: Water Use by Sector in Central Arizona[9]



Municipal Water Uses

Municipal water use in Central Arizona is roughly the same as it was twenty years ago, even though population has increased by around 2 million since that time.[10] There are many reasons why demand has held steady: a transition away from grass and water-intensive landscaping to desert landscaping, increases in water efficiency in many household appliances, the adoption of water efficient plumbing fixtures and increasing tap water costs – to name a few.

Figure 3: Municipal Water Demands and Population in Central Arizona[11]



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[9] *Id.*
[10] US Census, www.census.gov.
[11] ADWR, AMA Data

Water Sources by Sector

The *source* of water used to fulfill a particular need is as important as the total demand for that need. Water sources in Central Arizona include:

- local surface water supplies from the Agua Fria, Gila, Salt, Santa Cruz and Verde Rivers,
- Colorado River water imported through the CAP canal,
- reclaimed wastewater (effluent) and
- groundwater.

Surface water and effluent are regarded as renewable because their use to meet demand does not result in a permanent depletion of supply. In Central Arizona, surface water supplies are fully allocated and there are significant constraints on the transferability of these supplies from one user to another. Colorado River water imported through the CAP canal is a renewable water source, but the river is over-allocated and its flows have been diminishing due to a prolonged drought and aridification of the Colorado River basin watershed.[12] Effluent is also a renewable supply, and the entity (such as a city) that treats wastewater to produce it may put the water "to any reasonable use it sees fit."[13]

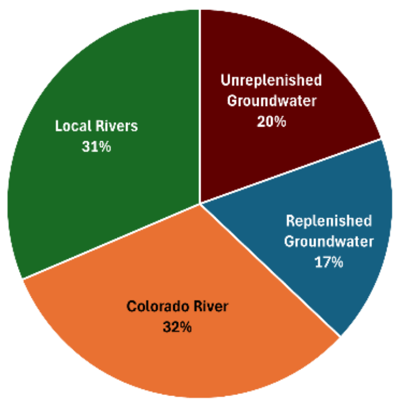
Groundwater is different. In Central Arizona and other parts of the state, only about 3% of the precipitation that reaches the ground converts to groundwater;[14] yet pumping technology enables the withdrawal of groundwater far faster than it can naturally be replenished. Accordingly, groundwater is regarded as nonrenewable, and managing groundwater as a finite source has been a defining water management challenge for Arizona since the mid-20th Century.[15]

Within Central Arizona, groundwater is highly regulated pursuant to the GMA and associated Assured Water Supply (AWS) rules, which generally promote the use of renewable water supplies and discourage the depletion of local groundwater.[16] However, groundwater has long served as a significant water supply for agriculture, and its continued use for crop irrigation is grandfathered under the GMA. This use is

[12] Bradley Udall & Jonathan Overpeck, *The twenty-first century Colorado River hot drought and implications for the future*, American Geophysical Union, February 17, 2017, https://agupubs.onlinelibrary.wiley.com/doi/full/10.1002/2016WR019638.
[13] *Arizona Public Serv. v. Long*, 160 Ariz. 429 (1989).
[14] Karem Abdelmohsen et al., *Declining Freshwater Availability in the Colorado River Basin Threatens Sustainability of its Critical Groundwater Supplies*, AGU Geophysical Research Letters (May 27, 2025) https://doi.org/10.1029/2025GL115593.
[15] Kyl Center for Water Policy, *Groundwater Protection in the Valley of the Sun,* ASU Morrison Inst., March, 2024, https://morrisoninstitute.asu.edu/groundwater-protection-valley-sun.
[16] For further information about the 1980 Groundwater Management Act and the Assured Water Supply rules, *see* Kathleen Ferris and Sarah Porter, *The Myth of Safe Yield*, Kyl Center for Water Policy at ASU Morrison Institute, May 2021, https://morrisoninstitute.asu.edu/sites/g/files/litvpz841/files/the_myth_of_safe-yield_0.pdf.

"unreplenished," meaning that it results in aquifer depletion, and it is the largest source of unreplenished groundwater demand in the AMAs. According to a 2020 ADWR report, agriculture was responsible for 63% of unreplenished groundwater pumping in Central Arizona from 2012 to 2016,[17] averaging around 892,000 acre-feet per year in the last decade.[18]

Figure 5: Tap Water Sources in Central Arizona



Largely due to AWS requirements, municipal tap water in Central Arizona is much less dependent on groundwater. In 2022, approximately 20% of municipal tap water supplies, around 215,000 acre-feet, derived from unreplenished groundwater pumping.[19]

Effluent is currently used exclusively for non-potable purposes, though a consortium of cities plans to treat it for potable purposes in the future. In 2022, about 75,000 acre-feet of effluent was delivered to the Palo Verde Nuclear Generating Station, nearly as much (70,000 acre-feet) was delivered to turf and landscape uses and another 70,000 acre-feet was delivered for agricultural uses. Approximately 70,000 acre-feet of effluent is recharged annually into local aquifers to replenish groundwater pumping and to build a store of water that can be used in the future.[20]

Part II. Water Sources for Non-Agricultural Uses

Sources of water available for non-agricultural uses in Central Arizona include:

- surface water and Colorado River water delivered directly from a canal,
- reclaimed water delivered via a reclaimed water pipeline,
- water from wells operated independently from the municipal water provider, and
- tap water delivered from a municipal water provider.

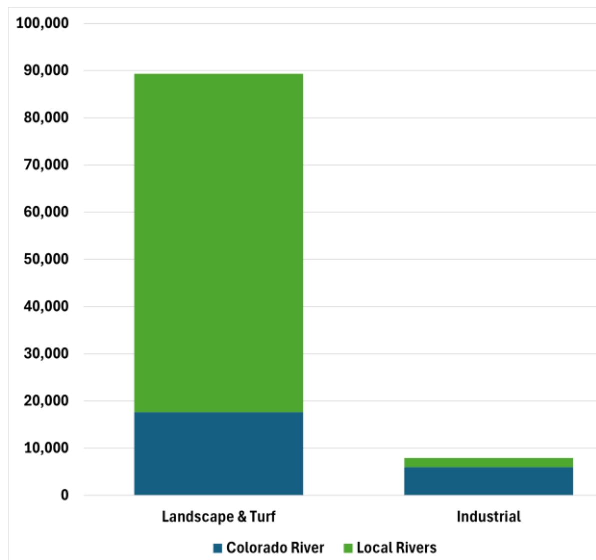
[17] Arizona Department of Water Resources, *Issue Brief: Unreplenished Groundwater Withdrawals,* June 15, 2020, https://www.azwater.gov/sites/default/files/2022-08/ISSUE%2520BRIEF%2520-%2520Unreplenished%2520-%2520Final.pdf.
[18] ADWR, AMA Data
[19] Kyl Center for Water Policy, *2022 Community Water Use Study,* ASU Morrison Institute.
[20] *Id*.

Surface Water and Colorado River Water Delivered from a Canal

Most businesses and industries in Central Arizona lack direct physical and legal access to surface water or Colorado River water; only a relatively few power generators and mining companies have access to Colorado River water imported through the CAP canal or to surface water supplies.

However, many turf facilities – golf courses, sports complexes, parks, cemeteries, schools and other places that grow grass – have access to surface water and Colorado River water delivered from a canal. Many turf facilities and some homeowners in the Phoenix area have access to Salt and Verde River water delivered via canals operated by the Salt River Valley Water Users' Association, commonly known as the Salt River Project (SRP), and most of the surface water available for turf and landscape irrigation comes from this local supply. In contrast, most of the surface water delivered for industrial purposes is Colorado River water. Nearly all of the Colorado River water delivered via the CAP canal for industrial purposes goes to mining.

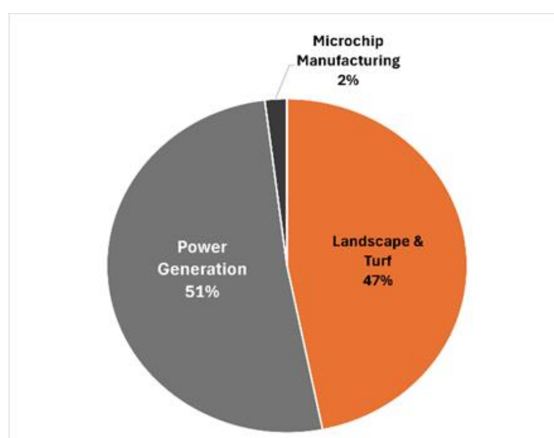
Figure 6: 2022 Colorado River and Surface Water Deliveries via Canal for Turf & Landscape and Industrial Purposes in Central Arizona (acre-feet)



Effluent

Effluent is also delivered to turf facilities in large quantities, and as noted previously a large amount of effluent is delivered for power generation. Most of the effluent delivered for power generation in Central Arizona goes to the Palo Verde Nuclear Generating Station but a small amount is also delivered to a natural gas generating station in Pinal County.

Figure 7: Effluent Deliveries in Central Arizona for Industrial Uses and Turf and Landscape Irrigation



TSMC WATER RECYCLING

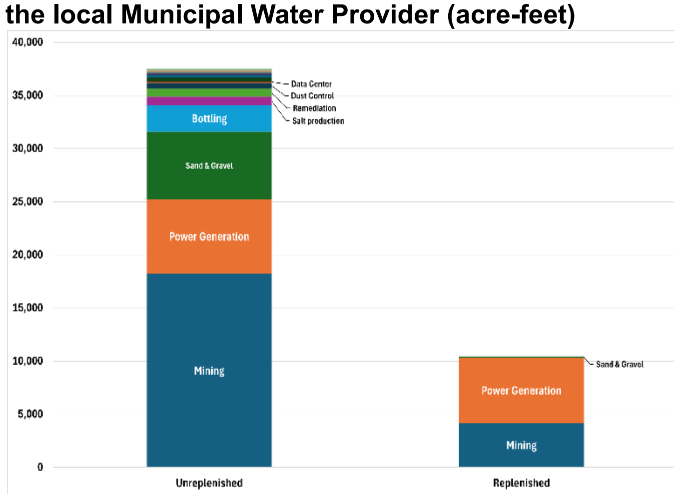
TSMC Arizona (Taiwan Semiconductor Manufacturing Company) in north Phoenix is a large industrial water user. While its first fabrication facility (fab) uses approximately 5,300 acre-feet of water per year,[21] the operation currently recycles 65% of that water which is reused in facilities support systems. TSMC Arizona is now building an [Industrial Reclamation Water Plant] (expected to begin operations in 2028) that will increase the water recycling rate to 85% and eventually as high as 90%.[22] This plant will allow the company to return water to an ultrapure state for reuse in its semiconductor wafer manufacturing process. Large industrial facilities are often capable of reusing water in ways others cannot.

TSMC is expected to invest $165 billion[23] in the regional economy—a comparatively favorable return on the water.

<u>Water from Wells Operated Independently from the Municipal Water Provider</u>

Under the GMA some large urban water users, including mining companies, power generators, general industries and turf facilities, are allowed to pump groundwater from wells that are not owned or operated by a municipal water provider. Often, these users rely on rights to pump groundwater that are independent of the municipal water provider and have no replenishment requirement,

Figure 8: Industrial Well Pumping Independent from the local Municipal Water Provider (acre-feet)



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[21] https://www.azcentral.com/story/money/business/tech/2024/11/04/phoenix-provides-water-to-a-new-chipmaker-any-cause-for-worry/75917812007/

[22] The company clarified in an email to AZ Central that this 90% goal applies to water remaining *after* evaporation occurs in the manufacturing process. Accounting for evaporative losses, the total water reuse goal is 74%. *'A thirsty operation': TSMC plant arrives amid water doubts, but Phoenix isn't worried* https://www.azcentral.com/story/money/business/tech/2024/11/04/phoenix-provides-water-to-a-new-chipmaker-any-cause-for-worry/75917812007/

[23] TSMC, *TSMC Intends to Expand Its Investment in the United States to US$165 Billion to Power the Future of AI*, March 4, 2025, https://pr.tsmc.com/english/news/3210.

meaning they deplete local groundwater stores. Golf courses and large turf facilities can own and operate wells independently from the local municipal water provider, but those developed after 1985 must replenish their groundwater use and are subject to strict conservation requirements.

Some enterprises avoid using tap water for financial reasons. Tap water delivered by a municipal water provider is typically more expensive than groundwater pumped independently from the municipal water provider or surface water or Colorado River water directly from a canal. Apart from costs for infrastructure and power to operate a pump, costs for groundwater include only the state's Water Quality Assurance Revolving Fund fees of $2.12 per acre-foot plus groundwater withdrawal fees (applicable in Active Management Areas) of $2.00 to $3.50 per acre-foot rather than the industrial class water rates charged by municipal water providers, which among the larger municipal water providers are on the order of $700 to $2,200 per acre-foot.

From the perspective of a large water user, there are pros and cons to receiving tap water service from a municipal water provider rather than using local groundwater. Tap water is extremely reliable. Municipal water providers use looped water distribution systems so that water service can continue even when there is a pipeline break. Tap water meets federal Safe Drinking Water Act (SDWA) requirements and can be used for commercial and industrial purposes and for potable needs such as in employee breakrooms and bathrooms. A large water user opting to drill and operate wells is responsible for developing and maintaining a water supply. Drilling and operating wells, and treating the water to meet SDWA standards if necessary, can be expensive and risky, but over time may be less expensive than paying the water tariffs of the local municipal water provider.

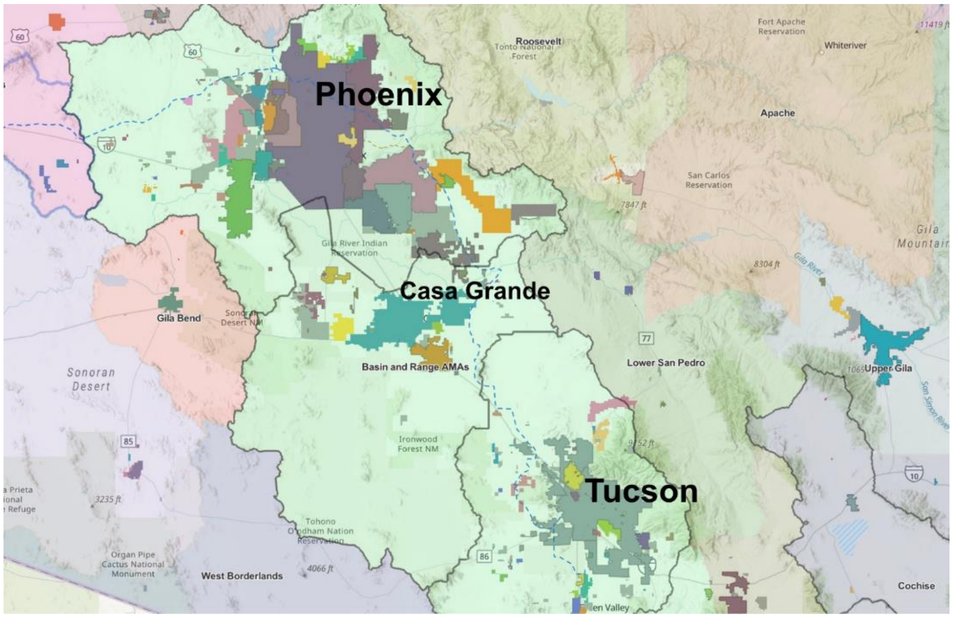
In some cases, the large water user has no choice but to apply to the local municipal water provider for service because well-spacing rules do not allow new wells to be developed on-site, or because local groundwater is not of adequate quality for the intended use among other reasons. In other cases, the large water user chooses to locate within the boundaries of a municipal provider because the user discharges a large amount of wastewater, and health codes and Clean Water Act requirements dictate that within and near city boundaries large discharges of wastewater be connected to an existing sewer and wastewater treatment system.

Water supply, reliability and quality are not the only considerations for a large water user when deciding whether to locate within the boundaries of a municipal water provider and to apply for tap water service. An enterprise may require a large and proximate labor pool, reliable power supplies, adequate street transportation and traffic control, solid waste service, housing and other amenities. These requirements may dictate location choice to a greater degree than considerations related to water.

Tap Water Provision

In Central Arizona the provision of water service for the vast majority of homes and businesses is undertaken by municipal water providers who can spread the large capital cost of delivery of safe, reliable drinking water across the customer base. These community water systems extend across all urbanized parts of Central Arizona.

Figure 9: Community Water Systems in Central Arizona



The majority of Central Arizona residents and businesses – some 4.3 million people and tens of thousands of commercial and industrial enterprises – rely on tap water provided within the service territory of a municipal water provider with a Designated Assured Water Supply (DAWS).

A DAWS is issued by the State of Arizona through ADWR. To achieve a DAWS, a municipal water provider must demonstrate it has the necessary water to meet for 100 years current and projected demand within the designation period. It must also show the financial capability to construct the delivery, storage and treatment works needed to make that water supply available to its customers. For a city or town, this means the adoption of a five-year capital improvement plan that provides for the construction of these works and certification by the chief financial officer that finances are available to implement the plan. ADWR must review a DAWS at least every 15 years to determine if it should be modified or revoked. This rolling time frame guarantees that commitments for water service do not exceed available supplies.[24]

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[24] Kathleen Ferris and Sarah Porter, *The Elusive Concept of an Assured Water Supply: The Role of CAGRD and Replenishment* at 7, Kyl Center for Water Policy at Morrison Institute, , Fall 2019 (citing Ariz. Rev. Stat. § 45-402(31); Ariz. Admin. Code R12-15-720 & R12-15-711).

Most DAWS providers' portfolios contain multiple sources of supply – some combination of renewable surface water supplies, reclaimed water and a quantum of groundwater, the use of which must be deemed consistent with the management goal for the AMA.[25]

If a municipal water provider has attained a DAWS, it has successfully demonstrated that its water management strategies are sound – renewable water supplies are secured, adequate finances support the utility, necessary infrastructure is planned and any groundwater pumping is consistent with the GMA.

To the extent that large water uses are served tap water from a municipal water provider with a DAWS, there is less cause for concern than if the large water use relies on the depletion of groundwater. Because a DAWS provider is required to continually demonstrate to ADWR that the provider can meet all of the demand within its service areas for the long-term, the DAWS provider is incentivized to carefully consider how a commitment to serve a high-volume water user will impact its ability to meet current and future demand.

Still, provision of service from a municipal water provider with a DAWS does not mean there is no concern regarding new large volume water uses. Current and future cuts in Colorado River water supplies imported through the CAP canal will put stress on the ability of municipal water providers to maintain their DAWS. Water managers, city councilmembers and local customers and constituents will need to carefully balance the benefits a proposed large volume use may bring in terms of tax revenues, jobs, and other considerations against the need to maintain a DAWS.

In some cases, new large volume water uses can be offset by the diminishing water demands of other customers. In other cases, the municipal water provider will need to reduce total demand or incur the expense of developing new water supplies to maintain its DAWS.

Part III. Large Water Uses within a Municipal Tap Water Distribution System

As detailed in a recent Kyl Center report, within Central Arizona, ten water providers serving a combined total of over four million residents have implemented ordinances or other measures regarding the provision of water service to high volume uses.[26] Generally, these measures provide a framework for assessing whether adding demand

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[25] Ariz. Rev. Stat. § 45-576.

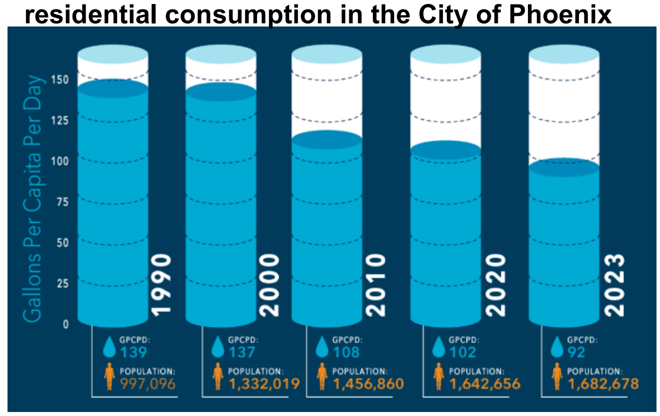
[26] Kyl Center for Water Policy, How Arizona Municipal Water Providers Are Regulating Large-Volume Users, ASU Morrison Inst. for public policy, September 2025, https://morrisoninstitute.asu.edu/sites/g/files/litvpz841/files/2025-09/Large-Volume%20Water%20User%20Ordinances%20090525_revised2.pdf. Of these ten providers, nine are city water service departments with DAWS. The sole private water provider, EPCOR, serves multiple service areas, three of which have DAWS: Chaparral (Fountain Hills), San Tan and West Valley.

from a high volume use is within the community's interests. Allocating a portion of a limited water portfolio represents an opportunity cost that should be carefully considered.

However, the demands municipal water providers meet are not stationary over time. Business enterprises come and go, technologies change, the water efficiency of appliances and industrial equipment increases, water service territories expand, population increases and turf and landscaping patterns change. Municipal water use per capita has been in decline over the last several decades in much of Central Arizona.

Because of this, an increase in the number of large water users served by a municipal water provider does not necessarily create an increase in the overall demand the provider must meet. Often, new water demands are offset by decreases in existing demands. In these cases, adding demand from a new large water user may contribute revenues from water sales to an otherwise declining revenue base.

Figure 10: Change in population and per capita residential consumption in the City of Phoenix



Source: City of Phoenix

As examples, total tap water deliveries between 2019 and 2024 fell in the city of Mesa, even though Mesa has the third-largest number of data center campuses in Central Arizona, and total tap water deliveries between 2019 and 2024 fell by 2,266 acre-feet in the city of Tucson, more than the amount of water proposed to be used annually by the controversial Project Blue data center and an associated campus.[27]

Large water uses have always been woven into the fabric of urban living. Golf courses and water parks are obvious, but manufacturing and other enterprises rely on water to some degree.

[27] Annual reports submitted by the cities of Mesa and Tucson to ADWR; Dylan Smith, "Tucson deal: 'Project Blue' data centers would thirst for water & electricity," AZMirror (July 18, 2025) (quantifying water demands for two associated data centers at 2,000 acre-feet per year).

Table 4: Estimated Ranges of Large Water Uses in Central Arizona by Facility[28]

Water Use	Acre-feet per Year
Power Generation	500-100,000
Semiconductor Manufacturing	5,000-20,000
Sand & Gravel Mining	100-3,000
Golf Courses	300-1,500
Beverage Facilities	100-1,200
Data Centers	150-1,040
Parks	5-700
Construction/Dust Control	30-500
Schools	5-150
Food Processing	10-100

This table shows estimated use per facility, not total use across each sector. Facilities vary widely in their water use based on size, production levels and other factors.

The largest industrial water user in Central Arizona is the Palo Verde Nuclear Generating Station, whose primary water source is renewable effluent. Of the groundwater pumped for power generation in Central Arizona, 47% is replenished with renewable water supplies.[29] This is not to say that the amount of water consumed for power producers doesn't matter, but rather to note that the source of water used for power production matters a great deal. To the extent that power production relies less on unreplenished groundwater, there is less concern.

WHY DON'T WE TALK ABOUT ALL THE WATER USED FOR BEER?

In 2024, bottling and beverage facilities used just over 3,800 acre-feet of groundwater in Central Arizona (not delivered via a tap water system). That's more than data centers use annually.

Why don't we hear consternation about this use of water? Possibly because a lot of people love craft beer. The popularity of a water use does not change its impact on local aquifers, but it may shape the public perception of how much it does.

Source: Thrillist & Arizona Wilderness DTPHX

[28] Kyl Center for Water Policy, 2022 Community Water Use Study, ASU Morrison Institute.
[29] *Id.*

Part IV. Considerations Related to Water Use by Data Centers in Central Arizona

How Data Centers Use Water

A data center is a building full of hard-working computer servers that generate heat and require climate control (cooling, in sunny Central Arizona). In the early days of data center development, the preferred cooling technology was evaporative cooling.[30]

> In an evaporative cooling system, pumps push cold water through pipes in the data center. The cold water absorbs the heat produced by the data center servers, turning into steam that is vented out of the facility. This system requires a constant supply of cold water.[31]

Traditional evaporative cooling entails a high consumptive use of water because the water that evaporates is not captured for re-use.[32]

Not all data centers use evaporative cooling, however. Some deploy electric cooling systems which require no onsite use of water. Some use a combination of evaporative and air cooling. And some use closed loop cooling, which is similar to evaporative cooling but "rather than venting steam to the air, air-cooled chillers cool down the hot water," which is recirculated through the cooling system, resulting in lower consumptive water use.[33]

The direct use of water for cooling is not the only water consumption by data centers. Running a warehouse full of servers requires significant electric power, and some types of power generation consume water. This water use is termed "indirect consumption." Most of the alternatives to evaporative cooling require more power than evaporative cooling. However, new cooling methods and technologies promise greater energy efficiency as well as lower water use.[34]

Water Use by Central Arizona Data Centers

It is difficult to quantify the exact number of data centers in Central Arizona, and even harder to quantify their water use. Data centers vary in size, are at different stages of

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[30] Peyton McCauley & Melissa Scanlan, *Data centers consume massive amounts of water – companies rarely tell the public exactly how much,* The Conversation, August 19, 2025, https://theconversation.com/data-centers-consume-massive-amounts-of-water-companies-rarely-tell-the-public-exactly-how-much-262901.

[31] *Id.*

[32] Arizona State University researchers and others are developing technologies to "harvest" evaporated water from cooling towers, thus decreasing the overall water consumption for evaporative cooling. Arizona State University's Global Center for Water Technology et al., *Third International Atmospheric Water Harvesting Summit*, January 15, 2026, https://asuevents.asu.edu/event/awhsummit2026.

[33] McCauley & Scanlan*, supra* note 30.

[34] Ram Sunkara, *The Evolution of Data Center Cooling: From Water to Emerging Technologies,* POWER Magazine, October 22, 2025, https://www.powermag.com/the-evolution-of-data-center-cooling-from-water-to-emerging-technologies/.

development or renovation and use a variety of cooling technologies. Complete information regarding cooling technologies deployed at every data center, power consumption and water use is not publicly available.

The AMA management plans include water efficiency and reporting requirements for all large-scale cooling facilities, regardless of whether they are served by a municipal provider or pump their own groundwater.[35] However, these requirements do not differentiate cooling towers for data centers from other cooling towers.

Combining three publicly available national data center maps and other validating data, the Kyl Center has identified 225 existing or planned data centers on 93 existing or planned campuses in Central Arizona as of the date of this publication.[36] Data center locations were validated by aerial maps, direct communication with relevant companies and publicly available documents, such as press releases and city and county meeting agendas and minutes.

This analysis finds that nearly all of the data centers identified receive water service from a municipal water provider. In the AMA Management Plans for the Phoenix, Pinal and Tucson AMAs, ADWR confirms this finding, stating, "Most large-scale cooling facilities are served by municipal water providers."[37]

Of the 93 data center campuses in the study area, 64 appear to have completed construction, 2 are under construction, and 27 are in a pre-construction "planning phase." The 66 data centers that are completed or under construction appear to receive tap water from a municipal water provider. We assume this is the case based on the data center location within the boundaries of the municipal water provider and the absence of industrial use permits, ADWR well records or other mandatory reports that would connect a data center's water use with groundwater pumping independent from the municipal water provider.[38]

With the exception of the city of Buckeye, all of the municipal water providers serving data centers have a DAWS. As explained in Part III, the AWS regulations mitigate water supply risks to existing users and aquifer depletion. When a DAWS provider commits part of its portfolio to a particular use, the result is a loss of future opportunity because

[35] These requirements apply to any facility "with a total combined cooling capacity greater than or equal to 1,000 tons. *See, e.g.,* ADWR, Fifth Management Plan: Phoenix Active Management Area 5-1101(23) (definition), 5-1110(A)(2) (applying large-scale cooling facility requirements to facilities served by municipal providers), 6.2.4 (large-scale cooling facility program description) (2022) https://www.azwater.gov/sites/default/files/media/Phx5MP.pdf.
[36] A single location, or campus, may hold two or more data centers.
[37] *See, e.g.,* ADWR, Fifth Management Plan: Phoenix Active Management Area 6.2.4 (2022) https://www.azwater.gov/sites/default/files/media/Phx5MP.pdf.
[38] A permit to withdraw groundwater to serve a data center is not available within the service area of a DAWS provider. ARS § 45-515.

that water is not available for a different use. By design, the AWS rules force DAWS providers to consider how adding new large-volume demand will impact the long-term water resilience of the provider's other water users.

Table 12: Data Center Campuses in Central Arizona

Water Provider	Built	Planned	Unreplenished Groundwater Pumping (% of 2024 Tap Water Deliveries)	DAWS	Large Water Use Ordinance
Avondale	0	3	0.40%	Y	Y
Buckeye	0	2	87.90%	N	N
Chandler	6	1	13.40%	Y	Y
Eloy	0	1	100%	Y	N
EPCOR Santan	0	1	85.70%	Y	Y
EPCOR West Valley	2	3	52.80%	Y	Y
Gilbert	1	0	6.20%	Y	Y
Goodyear	5	1	0%	Y	N
Marana	0	1	0.60%	Y	Y
Mesa	8	3	3.70%	Y	Y
Peoria	0	1	4.50%	Y	Y
Phoenix	25	5	0.90%	Y	Y
Scottsdale	5	0	9.50%	Y	N
Tempe	5	0	13.50%	Y	Y
Tucson	9	0	8.70%	Y	Y
Provider Undetermined	0	5	N/A	N/A	N/A
Total	66	27			

Phoenix, Tucson and Mesa operate the largest municipal water systems in Central Arizona and host the largest number of data centers. These providers rely primarily on renewable water supplies rather than unreplenished groundwater to meet tap water demand. For context, 1% of Phoenix's, 8.7% of Tucson's and 3.7% of Mesa's 2024 tap water deliveries derived from unreplenished groundwater.

One data center facility shares a well with Goodyear, the local municipal water provider. The water Goodyear pumps from the well was reported as replenished but the water the facility pumps from the well, 362.6 acre-feet in 2024, was reported as unreplenished groundwater.[39] An exhaustive search of ADWR databases did not uncover other instances of data centers relying on their own pump rights independent of the local municipal water provider. However, it is possible that other data center facilities have arrangements to use the same well as the local municipal water provider; if so,

[39] 2024 Annual Report to ADWR by right number 58-103222.0019.

municipal water providers appear to be reporting such groundwater use within their pump authorities and under the same AWS requirements as for all other customers.

In 2025, EPCOR received an Alternative Designation of Assured Water Supply (ADAWS) for EPCOR West Valley a combined group of its service areas in the western part of Greater Phoenix that includes 2 completed and 3 planned data centers. An ADAWS provider is allowed to “borrow” groundwater from the aquifer on a commitment to replenish it and permanently reduce unreplenished groundwater pumping within the service area over time as renewable supplies are secured.

The planned data center campus in Marana expects to receive industrial water from the Cortaro-Marana Irrigation District, not the town of Marana’s community water system.[40] Cortaro-Marana Irrigation District delivers a combination of groundwater and Santa Cruz River water to its customers. In order for the irrigation district to serve groundwater to the planned data center, the data center would have to acquire a grandfathered groundwater right or a general industrial use permit, neither of which require the groundwater pumped to be replenished.

A developer has announced plans to build a large technology campus, including several data centers, in the Pinal AMA west of Eloy and outside of the boundaries of any municipal water provider.[41] . We are unable to determine whether this facility will be served with unreplenished groundwater.

The city of Buckeye has announced plans for a large data center near the Buckeye Municipal Airport.[42] Buckeye does not have a DAWS or an ADAWS. Available public records do not specify the water source for the data center campus: possibilities include effluent and unreplenished groundwater.[43]

<u>Indirect Water Consumption by Central Arizona Data Centers</u>

Data centers require significant electric power. According to one estimate, by 2030 power demands for data centers in Arizona could increase as much as 165%, from

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[40] Jan Lesher, *Addendum Item Request for the December 16, 2025 Board of Supervisors Meeting* , Pima County Board of Supervisors, December 9, 2025, https://pima.legistar.com/View.ashx?M=F&ID=15028221&GUID=E5EA6EE3-DB51-4208-9E3D-3BB4A42A4B8E.

[41] Bex Staff, *State’s Largest Data Center Project Part of Vermaland Pinal Plan,* AZ BEX, August 1, 2025, https://azbex.com/planning-development/states-largest-data-center-project-part-of-vermaland-pinal-plan/.

[42] Tract Staff, *Tract Closes Acquisition of 2,069 acres in Buckeye, Arizona*, August 28, 2024, https://www.tract.com/news/tract-closes-acquisition-of-2069-acres-in-buckeye-arizona/

[43] Buckeye Tech Corridor Community Master Plan, approved by Buckeye City Council August 6, 2024 https://www.buckeyeaz.gov/home/showpublisheddocument/13925/638597547379700000

6,253,268 MWh/year to 16,602,465 MWh/year.[44] Power availability is currently a primary limiter on data center development in Central Arizona.[45]

Individual data centers are not required to report their power consumption and estimations risk gross inaccuracies, because the power demand for any single data center depends on multiple variables, including the weather, the size of the data center and the specific type of cooling technology employed, which may be replaced with more efficient technology during the life of the data center.

Moreover, the quantity of water consumed for power generation depends on the power generation source. Natural gas, coal and nuclear power plants use water for cooling. Hydro-electric power often entails storing water in reservoirs, which lose significant water to evaporation. Solar and wind power generation, in contrast, consumes virtually no water. And tracing power use to source is complicated. Arizona's large utilities deliver power from a diverse mix of generation sources and not all of that power is necessarily generated locally. Arizona's power utilities are connected to the Western Interconnection, which spans 14 US states and parts of Canada and Mexico.[46]

One metric for considering data centers' indirect water consumption is the relevant power utility's fleet-wide water use intensity (WUI), that is, the amount of water used to generate the utility's entire power portfolio, typically measured in gallons per MWh. The water use intensity of electric power production in Arizona has gradually declined with shift away from coal to natural gas and renewable energy.[47] SRP and Tucson Electric Power (TEP) have implemented goals related to reducing WUI.[48]

As stated previously in this report, the source of the water consumed is important. The use of renewable effluent or surface water for power production is not the same as the

[44] Electric Power Research Inst., *Powering Intelligence: Analyzing Artificial Intelligence and Data Center Energy Consumption*, at 13, May 2024, https://www.wpr.org/wp-content/uploads/2024/06/3002028905_Powering-Intelligence_-Analyzing-Artificial-Intelligence-and-Data-Center-Energy-Consumption.pdf.

[45] *See* Sasha Hupka and Clara Migoya, *Data centers guzzle Arizona's water and power. We calculated how much.*, *Arizona Republic,* February 4, 2026, ("Serving up the amount of energy that data centers require will entail a complete overhaul of the transmission system and adding lots more power generation. That's led to long waitlists for service in the Phoenix area – much to the frustration of data center operators.") https://www.azcentral.com/story/money/business/tech/2026/02/04/how-much-water-power-do-arizona-data-centers-use/87270206007/

[46] Federal Energy Regulatory Commission, Western Energy Markets Explainer (updated April 18, 2025) https://www.ferc.gov/OPP/western-markets-explainer.

[47] Kenneth Skinner et al., *Water Withdrawal and Consumption Trends for Thermoelectric-Power Plants in the Conterminous United States*, 2008-2020, ACS EST Water, 2025, https://pubs.acs.org/doi/10.1021/acsestwater.5c00360?ref=pdf.

[48] SRP, *FY25 Progress Update: 2035 Sustainability Goals* (goal of 30% reduction in baseline 2005 WUI by 2035) https://www.srpnet.com/assets/srpnet/pdf/grid-water-management/sustainability-environment/SustainabilityReport25.pdf; Tucson Electric Power 2023 Integrated Resource Plan (projecting 80% reduction in water usage by 2035) https://docs.tep.com/wp-content/uploads/2023-TEP-IRP.pdf.

unreplenished use of groundwater. APS and SRP have also implemented strategies to reduce aquifer depletion for power generation.

There is no indication that WUI or net water use for power generation has increased in response to data center development in Central Arizona. But additional data centers will require additional power resources, either from Arizona's utilities or generated by the data center itself.[49] It is possible that the total volume of water used for power generation in Arizona will increase as the economy evolves. It is also possible that some of that water will be unreplenished groundwater, either within the AMAs (pursuant to a grandfathered right or general industrial use permit) or from areas outside of the AMAs, where groundwater use is largely unregulated and therefore an open resource.

Community Opposition to Data Centers

"The reality is this: Project Blue will be built in the Tucson metropolitan area, regardless of what the City of Tucson decides."

--Nikki Lee, Ward 4 Councilmember, City of Tucson, August 5, 2025.[50]

Source: AZPM

These words may turn out to be prophetic. On August 6, 2025 City of Tucson councilmembers voted against approval of "Project Blue" a data center proposed to be built within the Pima County Southeast Employment and Logistics Center and served by the city's municipal water utility, Tucson Water. Local residents united against the project and ultimately council voted against its approval unanimously.

However, a few months later a memorandum of understanding for the project was approved by the Pima County Board of Supervisors. A permit to pump groundwater for an industrial purpose can be granted by ADWR if, among other conditions, the use is located within 3 miles of the exterior boundaries of the service area of a city, town or private water company, and the proposed industrial user has been denied service at customary rate.[51] The city's denial of water service opened the door for the project to

[49] Under ARS § 41-1519 a data center's purchase of "equipment necessary for the transformation, generation, distribution for management of electricity" is eligible for sales tax exemptions.
[50] Nikki Lee, *Ward 4 Newsletter Project Blue: What It Means to Say No,* City of Tucson, August 5, 2025, https://www.tucsonaz.gov/Government/Mayor-Council-and-City-Manager/City-Council-Wards/Ward-4/Ward-4-Newsletters/Ward-4-Newsletter?display=article&article_id=3eca028.
[51] ARS § 45-515(A)(4).

proceed by pumping unreplenished groundwater from the same aquifer that supplies Tucson residents, but without the concessions required by the city.

The revised project will likely use much less water than proposed in August, but the concessions required by the city would have ensured the original project was “net water positive.”[52] Project Blue had agreed to invest $100 million dollars to extend a reclaimed water line to the site to supply its industrial water needs.[53]

A community meeting was packed with opponents of Project Blue on Monday, Aug. 4 / Una reunión comunitaria estuvo repleta de opositores al Proyecto Blue el lunes 4 de agosto. Crédito: Michael McKisson Credit: Michael McKisson

“Project Blue includes a $100M investment to expand Tucson Water’s reclaimed water system, extending it from Mission Manor to the southeast Tucson economic development areas. This represents the largest public infrastructure project funded by a private developer in Tucson’s history. All reclaimed water infrastructure described will be funded, designed, and constructed by Project Blue, in partnership with Tucson Water, then delivered to Tucson Water to own and operate.”[54]

In rejecting Project Blue, Tucson gave up approximately $100 million in city tax revenue and $100 million in reclaimed water infrastructure that could have been leveraged to convert existing and future customers from potable to reclaimed water use. Tucson Water wishes to encourage use of its reclaimed water infrastructure to minimize use of groundwater. Tucson Water will avoid delivering over 1,000 acre-feet of water in the future but will also forego more than a million dollars in operating revenue each year. Councilmember Lee summed it up this way:

“Neither path is without tradeoffs. But the decision now is not whether Project Blue will exist. It is whether Tucson wants to be part of how it is done.”

Part V. Conclusion and recommendations

Most of Central Arizona’s completed data centers receive water from a DAWS provider. AWS rules applicable to DAWS providers incentivize the use of predominantly renewable water supplies and limit the use of unreplenished groundwater, ensure that

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[52] City of Tucson, *Project Blue: Developing Sustainable Digital Infrastructure in Tucson FACT SHEET*, July 2025, www.tucsonaz.gov/files/sharedassets/public/v/1/government/city-manager-office/documents/project-blue-updated-fact-sheet_250714.pdf
[53] *Id.*
[54] Id.

new water demand does not interfere with the water provider's ability to meet existing customers' demands and require continued improvements in water efficiency.

The Groundwater Management Act and Assured Water Supply rules have enabled Central Arizona to absorb large water users, including data centers, without significantly depleting groundwater reserves or imposing risk on existing customers. Colorado River shortages will change this. Officials with oversight of DAWS providers should carefully consider the impact of new large water uses on the ability to maintain a DAWS and requiring concessions, such as those requested by Tucson Water for Project Blue, to help protect groundwater reserves and ensure continued water resilience.

Outside of DAWS providers' service areas, under certain conditions an industrial user may obtain a permit to use unreplenished groundwater,[55] and several proposed data centers appear to intend to rely on unreplenished groundwater. Given current limitations on other forms of groundwater reliant growth, challenges to Colorado River supplies and heightened competition for alternative water supplies, any proposed large volume use of unreplenished groundwater deserves scrutiny.

For an industrial use permit to issue, the Director of ADWR must determine that the AMA management plan can continue to be adjusted to accommodate the additional use consistent with the AMA management goals.[56] Given the physical availability findings in the Phoenix and Pinal AMAs, special attention should be given to this condition.

> *"In Glendale, AZ, for example, we anticipate a 73% annual water reduction from its previous life as a rose farm. We also successfully decommissioned a 1M GDP well on this site to protect the local aquifer for the community's future, as we did not need it thanks to our closed-loop system."*
>
> *Aligned Data Centers*[57]

There will be shifts in water use as Arizona's economy evolves. New, large water uses can sometimes be offset from diminishing water uses by other customers and sectors. Where this is not the case, elected officials must think carefully about tradeoffs.

[55] ARS § 45-515.
[56] *Id.*
[57] Email communication (February 9, 2026) (on file with the Kyl Center).